

ActInf GuestStream #003.1 ~ Sarah Hashkes, "Active Inference, VR, and Ps

GuestStream | Sarah Hashkes | 1:26:49

hello and welcome to the active
inference lab
this is active inference guest stream
number 3.1
and we're really excited to be here
today with sarah hashkes
we are going to be talking about as the
title slide suggests
active inference vr and psychedelics so
sarah i'll leave it to you to just
introduce
the topic yourself and the ideas however
you want
and it sounds like we're going to be
having a first section of the
presentation
then a question and answer and then a
second section of the presentation and
then we're going to have a second
overall discussion
so with that kind of structure in mind
just uh introduce yourself
and please take it away hi everyone
so yeah we're going to be talking first
about
my research in vr inactive in france and
then
my research in psychedelics and how it
relates to a few issues about
active inference and we'll have a short
break in the middle of our questions
so uh yeah my name is sarah hashquist
i have a master's degree in cognitive
neuroscience
i opened the first vr lab in the motor

control department

at radboud um where i was researching

vr's effects on our brain body

connection within the predictive coding

framework

i am now ceo of radix motion we're a

startup that combines neuroscience and

immersive technology

and uh we're focusing on empowering the

psychedelic ecosystem

with embodied tech so yeah let's get

right to it

a lot of this is uh basically all of

this lectures

based on my research from academia it

was pretty nice to go back

and and reread all the things uh from

2017

but uh before i actually get into it i'd

like to

activate our active and friends and

a very short interactive experience so

wherever you are especially if you've

been sitting a lot i actually suggest

you stand up

um and hopefully you can see my hand on

the screen

and i'm gonna ask you to put your hand

in front of my screen

and just follow it for a few minutes

moments

how about do it with your mouse move

your mouse yes

okay we can do that too if you can see

the oh you oh i see you're not seeing my

face in this okay

yeah so follow the mouse uh for a few

seconds

uh and see if you can track this
so what just happened here so okay
i gave you a auditory
um instruction that hit your brain
and created a goal a top-down prediction
um that you were supposed to well you
didn't see my hand using my mouse
but uh basically that prediction was
that the position of your hand
to the mouth should be the same position
the difference should be zero
um now uh
your eyes were then giving you bottom up
sensory input
um that were telling you oh there's this
difference between
the mouse and the hand and
due to active inference wanting to
minimize this prediction error
your hand was moved towards the mouse so
this is a very simple example
of active inference and this follow me
um instruction and activity
is really quite basic for how we learn
motor skills as children and you can see
this in the way
kids play games if it's simon says
or you know patty cake and the way
children's
look to their adult figures to actually
copy them
and learn movement patterns and
just follow me example was part of the
research i was doing so
we'll get into that so
what exactly was i trying to do well
i was really hoping that if i could
receive

visual cues from inside of a body
of a professional mover i would be able
to
activate this active inference ability
of my brain
and actually improve my movements i have
a
background in martial arts and dance but
i'm actually
from childhood a very clumsy human that
took me really really i guess a lot
longer to learn motor skills and i was
looking for a way to hack this
and i kept dreaming oh if i could just
be inside the body of my martial arts
teacher
would i be able to punch like him would
my body learn to adapt and move faster
and more accurately
because i wouldn't be receiving this
prediction
that i am actually him so um
this is sort of where i started uh um
exploring how i could research this
now why would i even think it's possible
using vr
so part of the things about vr is that
it doesn't
just activate your visual sensation
uh and sensory input it's actually a
multimodal
integration between your visual
perception
and your proprioceptive auditory and
even haptic if you're
currently using uh like haptic
controllers
um in my experiment i this was like we

were using the sdk one of oculus
no haptic controllers i was connecting
them to
lab equipment to get positions of the
hands
um but we were still connecting visual
and proprioceptive sensations
meaning basically when you move your
neck around the world
is still moving the way you're expecting
it to move
so you're getting the same uh upper body
and neck activations of muscles of where
you are
in space along with your visual
perception and that activates
these higher brain areas and really uh
gives you this what we call the sense of
immerse immersiveness right immersion
you're in that place your brain believes
it's real even though you know parts of
you know that it's not
now vr really gives us a very unique
ability to influence the brain because
we can start controlling the
relationship
between the different senses in ways
that either fit
these previous top-down biases or don't
so we can uh this will really modulate
the amount of prediction error
that we're receiving and activating in a
person's brain
um and there's uh really a lot of
experiments out there that's
that have been done for instance with
anorexia patients
and people uh with pain issues that

really prove that
we can give a new sense of self
um to a person so when i talk about a
sense
of self what am i talking about from the
predictive coding perspective
it's the really really uh um great
uh paper by absentes curious
um that really i go about explaining
this concept
of minimal self being
the top-down prediction that just
explains the sensory correlations
between the different senses
and a great example is the rubber hand
experiment
so um if you don't know this
a hand is your hand is actually hidden
from you like
in the photo but it's receiving sensory
input
uh at the same frequency that you're
seeing a rubber hand
being simul stimulated now uh
researchers are a bit cruel with this
experiment they at some point take a
hammer and smash the rubber hand
and you can really see um and register
and record the physiological
measurements from the person and how
that person gets stressed
now neural research is actually showing
that this isn't
um the same for everybody there's some
people
that seem to be more susceptible to this
versus others that don't there's some
amount of suggestibility

and i guess maybe strength of what is
your own self
versus how mutable this top-down
prediction is
um in society might be actually quite
different and we'll talk about that
later
but generally this there is a very
strong ability for us to change our
sense of minimal self and we do this all
the time when we were closing
our predictive abilities uh become able
to just predict away the sensation
um unless you know you're somewhere on
their autistic spectrum and then
every little itch and and sock can drive
you crazy
um but also using tools uh
these abilities really extend what we
call our minimal self
so i was hoping that uh with vr
i'd be able to transfer myself from a
clumsy person
uh into somebody that can do movements
that are
much much more difficult and a much
faster time um
so um what did i actually do i needed to
find a task
that was known to be a hard task that i
could measure
right what is the difference here so i
went with a path that's called
bimanual interference so bimanual
interference
uh is basically our inability
to create different movements with
different hands

um and get an actual interference
pattern between them
um a very simple game which is what i
coded in my experiment is try
to make a circle in one hand and a
square with the other
right and just to try for a second
and what basically happens uh
is that you're you go between phases of
making circular movement
and phases of making square movement
so uh i set out to look at this task and
see
can i improve this task in people with
various vr
interventions so um
oops i don't know why this is finding
here but okay
so let's look at a little bit of the
experimental design exactly
of all the different things that i was
doing and i'm not going to go into
exactly all the analysis you can read
the paper because i want to talk more
about like um
the actual discussions and you know
what are the ideas that come from this
that we can implement
in our daily lives um but there were
basically
uh a few different conditions one
condition was
a follow me condition similar to what we
did at the beginning
of this lecture where you had an avatar
that was able to do these movements and
your instructor to follow the avatar
the other instruction was complete the

movements yourself
so there's a self um experiment
and there's a follow experiment now on
the follow experiment
we were actually uh discovered that we
needed to hide
the square and circle because the the
minute people were
uh already predicting that we're doing a
square and circle
they were not following as much so
when you were following you are doing
random follow movements
uh along with periodic small parts of
circles and squares which i then analyze
specifically
those parts um so that was one
just like i'm going to say sort of base
like baseline condition to see what is
the difference between
following someone versus doing uh these
movements yourself
and um the second line that you can see
on this design experiment
um was a bit of
making things um stranger with what we
can do as we are
so the second uh condition when you
moved your right hand you actually saw
your left hand move
now there's no way i can recreate this
for you without putting you inside of vr
it's a very trippy experience
everybody's like wait
what you're like moving your right hand
but your left hand
um is is moving uh and that really like
messes up uh your predictions now

the third experiment i put you inside of
the avatar that was able to do these
movements
and then told you follow the movements
or just
become this avatar and follow the
movements from within this avatar
so we actually manipulated uh two things
one is the high level task goal right
follow versus move yourself now
we assume that the self movement has a
very strong hyper prior
of allowing only one task this is
something that
um you know we're sort of um
meant to be doing naturalistically one
task at every given time we also have
this basal ganglia that's really
blocking more than one activation of
um a physiological uh motor
activity so that was one thing we
manipulated follow versus
um but there's do yourself and the
second manipulation was of the visual
feedback right
do you see your own hands moving do you
see your weird cross hands moving
or do you see an avatar's hands
moving from the perspective of your own
hands
um now um there's
again this the difference here when i
crossed
your hands really giving you lots of
prediction error
this is a prediction error that doesn't
fit the normal minimal self
activation and predictions that you have

versus when i'm putting you inside of an avatar
that's uh moving in the same way that you are trying to move
i'm actually not doing something that's against
this minimal self um prediction and we'll see how these things
um gave different results so let's go with the uh start with the
first thing that i was you know most hoping to get oh i'll just plop
myself into an avatar that can do these movements and i'll be able to do them
well that didn't work in fact we got uh the exact
opposite so when i say this increase the interference
it means that the people were less able to perform this difficult task
um they were having even more uh periods of both hands doing the same
thing either circle or square and not able to differentiate
uh each of them so um what actually likely happened here right
and and this is sort of a theory because um i was not looking at neural
correlates i was just looking at behavioral data
but um there is a general consensus in literature that the visual modality
at least for most of us is much stronger than proprioception
data
so it seems that we were getting this mismatch between the sensory input right

of i'm seeing my hands doing the thing
but i'm feeling my hand in a different
place
right now i was hoping that we that
would cause active inference
pulling your hand to where they're
supposed to be
but it seems that the actual actually
the opposite happened
and because when i i asked people they
were all
actually believing that they were doing
the task better
some of them actually thought they were
the avatar they didn't even realize that
this was a pre-programmed avatar
they thought it was them um and all of
them reported it to be
easier um so
but when i looked at the data all of
them were actually doing
uh much worse so it seemed that the
visual modality
caused the proprioception modality to
actually
uh lower and become you know less
accurate
and your brain decided when it has this
this
a higher prediction or there's a few
tactics it can do one is active
inference the other
is modulate uh one of the sense
modalities and it seems that that's what
it did just lowered the proprioception
stuck what stuck with the sensation from
the eyes which it could predict
and that's the way it sort of minimized

the prediction error

so this uh and we can talk a bit more in
the discussion about the meaning of this
and and how um what it means for
training

but this was again um you know
it was uh um against what i was hoping
and against you know my my
initial prediction but thinking about
this a bit deeper

it does make sense because accurate
feedback

of what you're actually doing seems to
be incredibly incredibly important for
learning any type of skill so giving
this

false feedback um of somebody doing the
movements

seems to not be the way to go uh around
actually teaching movements um an
experiment i would love to conduct in
the future is having both of these
as an overlay like have um
the avatar that's able to do the
movements but

they'll give you a physical um ability
to actually know where your hands are
and this is something that um
if people have vr they can actually
download um

my my vr app that's called miu which
lets you play with these things yourself
it's um

it basically records 3d movements and
then you can move around them and also
get haptic feedback and see where your
hands are

inside so i played around with that a

bit with like professional jugglers um creating movements and then me stepping inside and needing to follow their hands but also receiving the feedback of where my hands are through haptic feedback and visual perception um but i did not do an academic research on that yet and that i think could be really interesting uh because i think there might be something there because obviously the prediction error was increased meaning your brain needs to deal with this prediction error in some ways so there is a potential of still creating uh better updates and maybe activating this active inference ability but we need to not only trust the visual perception uh visual versus proprioception the the visual just wins and you don't feel where you are anymore at that same level of accuracy so that was um interesting um now let's look at the other two conditions of what i found so um in the other two conditions i found that when i actually gave people the totally incorrect things that were against their minimal self i crossed their hands they were moving their right hand seeing the left hand some things that they've never ever experienced before they were able to follow the avatar better

but the self-movement movements were worse

right and that actually does uh make sense because again we're interfering

with this self-level prediction and having to maintain this goal of doing a self-movement while your whole minimal self is just like totally wonky and doesn't know what's going on and has a much higher prediction error

um is the causing interference and um making you not able to do this task as well

but a follow task you don't actually need the goal is really different you don't actually uh

need to have a goal high-level goal-oriented

self um structure that's creating these predictions

you just need to really minimize this location and space versus that the other person

um they were actually doing better um and we'll talk a little bit about uh why why

that might have happened um yeah um so just one more finding before we go into a bit more of the discussion

is that there was actually um

um two very different strategies

might have been employed here as we see between the two different tasks right

when you're in the self task um

this higher prediction error that you need to deal with is

you can't just make yourself disappear
you need to
keep maintaining this goal so
maybe your brain created a weaker
prediction
right and then it would just like lower
that prediction error by
having a wider sort of um i guess and
there's a lot of between like stronger
weaker and wider
um my specific professor was very um in
the in the prediction predictive
coding scheme looking at things from
actually not a continuous gaussian
but more of a discrete level and a lot
of this is about like
detail the level of detail of your
prediction
so if you have a less detailed
prediction you're going to decrease your
prediction error
right if you say and we'll talk a little
bit more about that when we talk about
psychedelics because it's very relevant
there
but if you're just saying okay my hand
needs to be
around here you know versus your hand
needs to be exactly
at a specific point that's another way
to decrease prediction error
so on the self task that might be what
was happening
your brain was just decreasing the
detail
of the prediction in order to decrease a
prediction error
which would result in worse behavior

versus the follow-up task your brain might have reduced the prediction uh error by actually reducing the visual perception and getting on the relative right between proprioception to visual actually getting more of this proprioception creating these closer feedback loops with where your body is in space and getting this more independence between your hands because you don't need to have this single task anymore you're just following um and uh an interesting finding i don't have the quote of the paper here but um there's a paper that did the exact same task but with the eyes closed and just touch so people had their eyes closed and were just touching uh something that was moving in a bi-manual independent manner and they were all able to do it and suddenly from becoming a very difficult task became a very easy task so um this really shows that sometimes and i think anybody that you know practices movement is it's sort of a common trope but there's a thing of like thinking too much so goal oriented movement can actually um if we are taking this goal oriented movement and putting it in a way that's too

detailed we're telling
our brains create these movement paths
and create two separate movement paths
that's a really not naturalistic state
for our brain to be in
um versus when we're just like okay
we're following things
that's a very naturalistic the low level
feedback between our motor cortex and
our proprioception and our sensation
can just work um
so yeah that that that's all sort of
hopefully understood um
now another finding that that happened
was that
um there was a much higher variance
like i was expecting this to be a very
hard task for everyone that's what the
literature said
um but i found that uh and again this is
a small experiment we're talking about
like 13
psych students right um uh all healthy
subjects
um young but uh
two of them were very able to do the
baseline task
and that was a surprise some of them the
task was so difficult
that i could barely use the data because
they were literally just moving one hand
the other hand was just freezing in
space and the other hand would move
so they weren't even able to maintain
these
two goals of move two hands um
in separate ways uh at all um
so uh that really goes to show like just

how
different our brains are one of the
interesting things to compare this
experiment to is a binocular rivalry
um which has been written a lot about in
the predictive coding literature
um so for whoever doesn't know this
binocular rivalry
is when we project two different images
onto
each eye so one eye is seeing one image
you can see in the picture
the right eye is getting a red image
with one
angle of direction and the left eye is
getting a different one
now what happens to your brain is it
actually
switches your perception switches
between seeing
one of the images and the other images
with
some short phases usually of like mixed
perception but really just bouncing
mostly between one to the next
now here too there's a lot of variants
in the population
and one of the interesting things is
that people on this
uh autistic spectrum which you know i i
have endless criticism about the dsm and
how
we're classifying uh autism and
generally mental health and different
types of um
brain brain abnormalities
um but some people
definitely have much more um

mixed states than others and different
drugs can also affect the ability to
have mixed states
and it's sort of interesting to compare
these two tasks because
both of them are sort of creating a
bi-stable system
coming from an unnaturalistic task
or a naturalistic um um input that
you're not
usually supposed to be doing because the
way our brain was really
meant to be doing motor actions is
around
high level goals completing high level
goals right
give me the border um turn
on the thing grab bang
there's always a high level goal that's
oriented towards
um one of our high higher level needs or
lower level needs but like a specific
need that you're trying to complete
um and with this
task of telling people oh move your hand
in this motor path's way and your other
hand in this motor pathway
it's very unnaturalistic for our brain
very similar
to giving someone uh these two different
visuals which we never see
in real life so um
again this is really correlated to this
having a very strong hyper prior in
in a binocular rivalry situation your
brain has a very strong hyper prior no i
can just be seeing one thing
this is this is how the world works

there can only be one thing
in one place and then your brain
activates
in order to minimize this prediction
error it goes between okay
i'm i guess i'm seeing this one now it
keeps getting prediction error
oh i guess i'm seeing the other one so
it keeps jumping between
these two states in order to minimize
the prediction error
now the same thing it quite likely is
happening
with this uh bimanual interference with
having a very
strong hype of prior of oh i'm only
supposed to be doing
one task at any given time you're just
giving me
two different tasks to complete at the
same time
i'm going to be switching between them
now possibly when you have this
different sense of minimal stealth
this can actually create different
activation patterns and let you be more
in a mixed state more in an independent
state
have basically weaker hyper priors
around
either what you're seeing or
what your task abilities are so this
might be an interesting way of generally
like understanding
um a lot of what our hyper priors are
doing to us
and just the variance in the population
might be around these hyper priors

so let's let's now let's get into like
more like
what are these possible implications um
and and how can we use this data
to like you know do cool things with
rehabilitation and even just training
and getting me to be less clumsy so
one of the things that comes up from
this is really naturalistic
design of movement goals instead of
telling people
you know move the head your hand at this
angle reach
over here um and we see this a lot
actually in
in traditional martial arts they give
movements names of activities
like in kung fu you have stroking the
horse the
the host of maine and karate we have a
move that's called
breaking breaking the wall or breaking
the gate
so having these um goals that are really
oriented towards
task completion versus
goals that are too low level
are much more likely to activate this
active inference in ways that we want
them to
when we put the goals into two
low-level um patterns um
we're then creating this sort of
unnaturalistic
um activity and maybe even like
slowing things down like the the fast
parts of our brain that deal with
bringing together these sense modalities

of visual sensation and
motor actions they need to stay down
there
they need to not be part of this high
level
self um
you know network uh part of our
hierarchy called network
and i think that's really interesting
because it really
is like what i was saying every martial
arts teacher
my whole life has told me stop thinking
and i like i don't know what to do about
this you know how can you stop
thinking um but part of it is
uh letting these low level parts of your
body
do the movements while maintaining high
level
goals that are not oriented to exactly
oh where's my elbow is it here or here
um
where am i holding my fist like all
these little details
uh work much better if you give them
high level metaphors somehow
that your brain that the lower levels
can then
take those predictions and minimize
prediction error
so um another interesting
part of this is that it might actually
explain uh
some of why mirrorbox therapy works
um and and why mirror rocket therapy is
now something that's also being used in
vr

and so for whoever doesn't know
mirrorbox therapy
is used on amputees and people that have
pain in one hand um
like chronic pain or even a stroke
people that have suffered through a
stroke so what you do is you take the
body part that is actually okay
that doesn't have pain that is working
and you use a mirror
to then get your brain to believe
that your other body part is also moving
and is healthy or even there
if it's an amputee and this was actually
uh figured out by i think the first
person was ramachandran
uh working with amputees that were
suffering from just phantom pain
and how do you get rid of pain from a
limb that doesn't exist anymore right
this was like a top-down prediction
from the injury that was just not
getting updated
um and you see this a lot in chronic
pain
really you see the neural circuitry even
change
as the predictions for pain just cement
themselves
and become biases in your brain
so um by using this type of
mirror box therapy um it
helps relieve the pain so why or how is
this connected to what we just saw
because the visual modality is stronger
than the proprioception modality
it's not too far fetched to believe that
the visual modality is also stronger

than
the gnostic scepters the pain receptors
that we're getting
so if we can actually cause your brain
to believe and and
just by seeing cause that visual
perception of a healthy hand
it might be enough to mute the same way
your proprioception was muting
to mute your nasty scepters um
and then reduce the pain so that's
that's really interesting and it's
interesting to see they are also
starting
to use this with stroke victims to
actually help them rehabilitate their
hand
um and that's a really interesting
because right that's a little bit
not in accordance with what i saw um
from my experiment
but it might be different because they
might not be getting
um current data about the hand
i think it might be really dependent on
you know what the stroke has actually
damaged in your brain
if the stroke part has damaged your
ability to receive any type of
proprioceptive data
um then maybe
getting that visual perception uh
might help reprogram that there might be
less of an interference going on i'm not
sure it's sort of interesting to think
about
um so let's see what are we left with oh
yeah

so now there's basically that third
uh condition why i gave this crossed
hands condition
and i think that's really an
underutilized
mechanism generally in any type of
rehabilitation and training
um messing your brain up even more
in some ways and i think that that's
a very strong intervention that we could
play around with a lot
more uh because it's you know 100
percent we're creating more prediction
error that your brain needs to deal with
now how your brain will deal with that
prediction error will it actually
uh remodel whether create active
inference
will it reduce some other sensation
might really depend on the task
on your brain um on how we do this but
being able to increase prediction error
in your brain is a very powerful ability
and i think that's something that really
is worth looking into more
when we're talking about rehabilitation
tools um
and we see these types of things
starting to emerge in here more with
like
total body replacements like i was
talking about about like um
anorexia patients uh putting them in a
whole new avatar and creating a full
body
um rubber hand experiment so
they're getting sensations on their body
at the same time that they're seeing

uh their avatar getting sensations and
it's
been proven to help them remodel this
top
level uh belief and accurate belief that
they have about their bodies being
overweight
even though um they're not so
that's very interesting but i think
there's really a lot that we can do
with um increase massively increasing
prediction error in the brain and you
see this in a lot of crazy experiments
right like
the upside down glasses experiment that
people
just within a short time were able to
totally adapt
and live in a world where everything is
upside down um
or with like training people on bikes
that work in a totally different
direction so
um sometimes by making things harder
we're making things easier um for our
brain
or we're forcing it out of this um
minimal potential well that it might
fall into
it's like okay now there's too much
prediction error it's time to actually
change
uh because you know as we've been
talking about like
um this minimal uh
uh prediction error if if you're in a
stable state where you've minimized
prediction error

that's not always a healthy thing for
brains

um and side note this is also
a model of depression basically not
having enough prediction error in your
brain

falling into a too stable state a low
entropy state of your brain

so increasing prediction error in many
ways can be very important for
our physical health our mental health
and it's something that i personally
try to hack on myself all the time i'll
do like

dark showers so i can't see anything uh
i'll try to use my left hand

to to to use toothbrushes just like very
basic daily activities that you get very
used to doing

just put another constraint on yourself
um and make it like a fun game it's also
a great way to train

in any type of physical task uh

okay so uh yeah we talked a lot about
uh the visual dominating their
proprioceptive but why

why is that so there's a paper from 2009

i'm not gonna i don't know how to
pronounce that you

um but uh their theory was that it's
around using

tools that in order for us to be able to
use

tools we need to be able to

uh not trust the proprioceptive

sensation as much as the visual

perception because we don't have data

from any proprioceptive data of any tool

that we're using
um so that's a very interesting theory
um but for my paper i think there might
be something actually
deeper than that going on i also think
and this might be interesting to
figure out a way to test this on animals
that don't really use tools
um but i would expect it to
uh possibly and this might might be very
different because some animals you know
actually smells their smell world is
much more dominant
their their hearing um but
i think that for this goal-oriented
movement where we need to coordinate
many body parts you know you're driving
a car
uh you're just even getting out of bed
and
um getting dressed or walking all these
things require
um lots of body movements working
together
so in order for that to happen
we really need to reduce the
proprioceptive sensation
so we can get this high level goal
to dictate the lower level hierarchies
of the brain to actually pull towards
the active inference
if we had very strong perceptions of
where we are
right now we wouldn't be able to move
into the next state of where we want to
be
so we're taking our brain is taking this
high level goal and breaking it

in to movement predictions
and for those movement predictions to be
able to
move your body into where we want it to
be using active inference
we need to weaken where you are
currently
um and have a a weaker now in order to
get to the future
um of this goal-oriented movement
that's that's sort of how i see it at
least
um a sort of interesting thing that i'd
like
to talk about and we can also open this
to discussions and questions
at the end is how this uh connects to
flow state
and improv and dance and play and all
these things that i'm you know totally
infatuated with
and want to spend most of my time doing
um so flow states uh for
if people haven't heard this term before
it's
um a term that's been coined when
uh athletes or really anybody that's
good at a specific task
is at their peak uh it's
been sort of defined as being freed from
self-consciousness
now a way to look at it through this
lens of the these experiments and these
hierarchical levels and predictive
coding and active inference
is um that it's basically
inhibiting actually these higher level
abstract goals and letting us

get hyper accuracy within this
proprioception modality
so really what we call being in the now
right getting incredibly fast feedback
loops
that aren't about the end goal it's not
about if you
i think talk to any of these athletes
it's not about winning
it's not about getting to the end it's
just like this pure
joy about being in the moment
um and and letting your body be
right um and
a lot of this uh really might have to do
with
inhibiting this higher level of our
neural hierarchy
so we can get this hyper accuracy of
proprioception so we can create these
fast feedback loops between our body and
the environment
to get to this like higher level of
performance
um and this takes us to more of like
improv dance
and play that are really non-goal
oriented
right this is sort of the definition and
how i see you know there's a big
difference between
games and play games there's always like
you're winning you're going towards a
thing or you're losing
um but in play there's none of that
there's just exploration um the same
goes really with improv dance compared
to

i don't know ballet or choreographed
dance you're just
exploring um the possibility space of
what you can do
um and that at least for me and i think
well not only for me this is quite
widespread anybody that is into these
things i think
finds them the most joyous um and you
know kids find them the most joyous
and it's a very very healthy part for
our brain
to not be in a goal-oriented
um movement pattern and to just
let go and explore without having
um a high level active inference
happening active inference is it's still
happening but it's being sort of
driven more by the bottom up signals and
really the moment to movement
possibilities without having this like
okay
let's gonna do let's do like a pretty
move let's do a pirouette let's do
something um
yeah so i'm i don't think i actually
want to show this video right now but
it's more about focused on vr but
if you want to look at it it's a cool
video that's called this is your brain
on vr that
uh we made um just
predicting the sound and stuff won't
work uh but it sums up a bit about these
what we've been talking about uh how
different modalities
get expressed uh activated in your brain
on we are

reprogramming your predictions and how
we can play with these things
okay so yep this this one this was
it that's awesome
wow i wrote down um a ton of stuff
so shall i ask uh some questions and
we'll head into another part
many places to jump in i really just
caught on to this weaker now like a
weaker current moment in order to get
further so that the way you connected
that to
accessible experiences like the play and
the martial arts really awesome
so when you talked about active
inference kicking in
what what is happening there and what
what did you mean by
active inference as far as um the way
that an individual person might relate
to it
uh i'm in what like domain all right
give me more
you talked about how you talked about
predictive coding and how active
inference would kick in
um or would go into a different mode
depending on whether a task
is being selected at a high level or a
low level or lower level
yeah exactly like we're always gonna be
i mean there's always prediction error
in our brain right
uh depending on what's happening now
are we gonna minimize it with with what
there's again these three different
tactics that your brain can implore
one um reducing the sensations

two updating your prediction three doing
an
action um so when you're just
in this free play
there's still a need to create a
prediction of your movement pattern for
it to actually happen
right but it's happening at a much lower
level of your
neural hierarchy it's not happening
because
there's a high level part that's saying
oh i want to look pretty
i want to get my teacher to like me so
i'll do something
um it's happening at a level of
the sensation uh at the level of what
feels good from the proprioception the
haptic
um the the pure like your body
knows what's joyous to your body or
these lower level
um predictions and biases that your body
has about
what is joyous healthy for your body the
knowledge
is there there are these lower level
biases
so we're allowing our system to collapse
into a state that's being
led by these lower level biases
um and this goes i think like in both
ways this can go very much into like
your body knows and has biases around
what's healthy and fun
and feels good but there's also that's
where trauma is stored too
there's an amazing book i think it's

called the your your the body remembers
and both emotional and physical trauma
really
are stored in our low-level biases
of our prediction around our movements
and our body
um that too can be you know a big
issue for many many people that need to
deal with this
and and being able to
go from a state where your brain's
triggered right and in the mental health
spaces it's called like when you're
when you've been triggered um and these
by these negative biases appear that are
predicting oh
pains how gonna happen even though
there's no objective reason for that
except
your history something's
triggered that prediction so being able
to switch
into that place
in your brain that knows what feels good
for your body
and what's healthy for your body is
super important and we don't really know
how to do that
in a very reliable way yet you know
there's mindfulness practices
breathing and stretching and massage
um but i think like my biggest fear is
that we're
with technology we're actually being
pulled away from our body
and that's generally what i'm trying to
close that gap um wow
that makes any sense thanks for this

answer and it's actually related to this
next question
you talked about how when the avatar was
getting almost like
false perfect feedback it didn't improve
the actual performance
so it really highlighted how the
technology can't just
um just simply mechanically demonstrate
like a scaffold it has to give
somehow accurate feedback yet it's
often fantastical scenarios or abstract
scenarios so
i wondered what were your thoughts for
technology
design how could technology be used to
actually
enable and enable these kinds of states
that you're
getting towards um yeah
without constraining us in that way yeah
so i think this was a very big learning
for me right
you need to give a person accurate
feedback of
where they are to and what they are
actually doing
um and there again there might be an
ability to
uh impose two such uh uh predict
things like you might be able to still
be in a first person perspective
of somebody doing the movements but also
either feel with vibrations like if you
try new that's what happens i just
use the haptic feedback to tell you if
you're are in the right place
you'll get a vibration so i'm

strengthening your ability to know where
you actually are so there's
sort of ways to hack our other sense
modalities
with you know music uh really it's a bit
of like gamification when you're on the
path
um of of doing the things that were
you're trying to learn uh activate the
this the even like the dopamine um
you know dopamine at least within the
predictive coding framework it's really
like a
precision signal for a for prediction
error so like
uh give some more bling bling bling with
a little bit
randomness when you are doing the things
that you're trying to learn
correctly but you need to get that
feedback
of when you're doing things correctly
um yeah and i think that that's what was
missing from sorry from this experiment
was uh giving and the faster and tighter
these feedback loops are the more your
brain will
perceive them and actually activate them
so there's some really great you know
meditation tools that you'll meditate
and afterwards you'll see your heart
rate um
there's some more professional like
biofeedback rhythms that you can see
what's happening
either with uh eeg or
um heart rate in real time
um and that's also actually i talked to

you before we started like i'm
build i'm starting my first hardware
project
um to do that and give a real-time bio
feedback of heart rate
with leds if you want to go it's called
the wisdom truffle
go to wisdomtruffle.com uh you'll see a
trippy truffle
and basically that's the idea of helping
people
reach some of these um tighter feedback
loops on a lower level
by hacking their their visual modalities
like let's use it we figured out okay
the visual modality is super strong
can we connect it to other sensations
that we don't have
as much insight to like could i connect
the visual modality to heart rate yeah
pretty easy could i connect the visual
mentality to perception
yeah if you go to mu when you dance and
move
your actual your avatar changes color
and the sparkles
change color so um by hacking
uh the the visual modality that really
is like our
brains graphic computer it's like super
strong
it's dominant can we use it in different
ways to connect
sensations that we don't have as strong
and are not as dominant in us that's
that's one of the things
awesome and that point you made really
um

about the high level priors like both my eyes

are looking at the same thing and you can kind of play with that by pushing on one lightly and you know it just does

distort the image and it's sort of

it just demonstrates uh it's one of

those examples like the high acuity in

the periphery or color vision the

periphery or the blind spot

one of those examples that help bridge

between

for example input output information

processing

understandings of cognitive sciences

versus this really

extended and predictive way that you're

talking about things so it's really

awesome and thanks for making it really

accessible with this presentation

so you want to go to the second part and

then if anyone has questions

live otherwise i'll just keep on writing

cool things down thank you

so okay

context

there's a counting switch um but we're

still staying within the predictive

coding framework and active inference

but um yeah i'm gonna just give you know

i really like this idea of giving visual

metaphors

based on all the things that i talked

about i think it's a great way to learn

and and have this intuitive

understanding

uh both of math i don't know if you

follow the three blue one brown videos

but
they're just phenomenal and and
mathematical understanding because it's
all
visual coded math but i'm going to give
a
visual metaphor you can either look at
the images or even close your eyes
and just imagine that when your brain is
creating reality it's very similar to
playing in a sand with some buckets and
making sand castles
and the sand is just this constant
incoming
bottom-up signal there's just so much of
it constant sand is just flooding
into your room into your space and you
have these buckets
and you're trying to take all the sand
and like fit them into the bucket
constantly to create
this um reality really and this is
pretty much what your brain is doing
with with uh
at least in the predictive coding
framework with the buckets being these
top top-down
priors and the sand being the bottom-up
and incoming signals
so there's a constant buckets trying to
predict and put the incoming signals
into a specific bucket that then turns
off
the sand basically now let's talk a bit
about the classical psychedelics
psychedelics they're starting to expand
to be
you know like basically everything but

um classical psychedelics are whatever
activates the 5-HT_{2A} receptor
agonist just means activating um
so um what really happens when
we take a classical psychedelic is that
these your brains
buckets are getting broken down
into smaller and differently shaped
buckets
so your priors are getting broken down
they're getting diffused
into uh sort of
weaker but more accurate um
or more detailed um uh buckets
that let your brain create an actual
different reality for itself now
uh where is this coming from uh when we
look at the actual
micro circuit of the brain um and this
is of the cortical
uh you have you know seven layers of
your cortex
and how the neurons are actually laid
out there this is a very amazing
paper by pasta we can see
exactly the feedback loops that are
correlated to
uh predictive coding where are the um
where's the data going upwards of which
we
see in these forward connections in
purple it's coming in from layer four
it's going up to layer
two and then looping to layer five and
then where are the backwards connections
that are coming from these higher level
brain areas that are correlated to our
predictions

um and we see that in green um
coming from layer five going down to
layer four
doing these these feedback loops um
so when we look at where the 5ht2a
receptor is
we see that they they're situated very
specifically
on these layer five um
on the backwards connections basically
so we're seeing that they're activating
uh hyperactivating and making the
sensitivity
of these neurons uh much more sensitive
so
that's that's why um the theory is that
it's causing these predictions to fire
at a much lower threshold if beforehand
let's imagine a whole neuronal
population
you'd need to get you know x amount of
neurons resonating together
um to create a prediction now you need
much less of them
just a few neurons are like throwing out
their own prediction
and a few other neurons are throwing out
their prediction and you're getting
many more competing predictions so
very simple uh metaphor if you're
walking down a forest
in a usual state you might have a
probability of 0.4 percent oh i'm gonna
see some animals
and a 0.6 percent prediction of i'm
gonna see some plants
but after you've taken psychedelics and
these predictions have

decomposed um your you might get a
much more detailed uh uh
prediction possibility space you know
maybe you're gonna see a dog a cat hey
maybe you're gonna see an elf that's you
know a low probability but it's
suddenly there so things that were uh
below the threshold of probability are
now becoming
a higher threshold uh and part of your
predictive
probability space now
what really happens here right um and
and what's really uh
sort of important is how this correlates
with
the rest of your bottom up sensations
um if you are in a very uh
getting very clean bottom up signal
which
almost never happens but let's imagine a
very clear
auditory signal um at least for a moment
or very clear
visual perception that you're getting
suddenly you can have
much higher accurate predictions
uh and this is uh you know people on
psychedelics
including myself will report this quite
often like oh wow
i can see footprints in the forest the
way i've never seen them before
or this perception of uh hearing music
and like the audio is just so crisp and
so detailed
in ways that i don't perceive them when
i'm not on psychedelics

um but if you are uh
in a bottom-up environment where you're
getting noisy signals
it's dark things are constantly changing
there's
various sounds um suddenly the the
possibilities and the predictions that
you might be getting
are might not be as correlated to
you know quote-unquote objective reality
suddenly your predictions
can become much stronger uh um
and just or much less you know
accurate but still a type of overfitting
might happen
that's i think a good word to use here
where you know oh
i'm in this forest it's dark and gloomy
there's something moving
oh it's an elf right because that was
part of the probability space of your
prediction
and with the noisy bottom up incoming
sensation that was enough
to reduce prediction error um but what's
important to note and you can see sort
of
in in this very uh simple image
that the decomposed predictions are all
uh less strong so they're not going to
turn off as much
prediction error so your your brain is
going to constantly be while you're on
psychedelics
in a state that has higher prediction
error which
as we saw before and we've been talking
about when you have a higher prediction

area your brain

has a few different tactics that it can

use to deal with it

one modulating the senses two

updating predictions three active

inference right

so this is why uh psychedelics offer a

very very unique opportunity of

updating your model because there's a

much higher level of prediction error

and then um under some situations your

brain might decide to actually

update the model and if people are

suffering um from trauma and depression

we're seeing

absolutely amazing results in research

uh around using psychedelics to update

people's biases around these things

um again very much depending on the

setting

don't expect to take psychedelics and go

to a place where you don't feel safe in

and you don't have a support structure

and to get these benefits

in order to update your model you need

to be getting bottom up sensory inputs

that are

very strong and accurate around what's

happening

to you right now but if you do that

and you are able to feel safe in your

body like we've talked before be in

place situations or improv or flow

that might allow your brain um to really

really heal

deep traumas to really reprogram deep

biases from childhood

when your brain was more uh had this

extra neural plasticity
so um now let's get to active inference
so the interesting part about these five
ht2a receptors
it is that we don't have so many of them
in our motor cortex there's a lot less
of them
so we can actually uh say that active
infra
inference modulates psychedelic
experiences and makes them
uh weaker and we see this both in the
instructions
that people are given when they're doing
psychedelic therapy
uh right now in um at maps for instance
or other
um universities and hospitals that are
researching this
they really tell people lie down don't
move
and when i talk to therapists about this
they could really say
we know when something's bubbling up
with the person we see this
in their body when it's becoming too
much they start really
moving and we try to encourage them to
actually stay still
because they they when we start moving
we're
actually weakening the prediction error
right our brain is creating a prediction
that's
still pretty stable because we don't
have as many
5-hc2a receptors that have been
activated

hyper-activated um because we took
psychedelics in the motor cortex
so our motor predictions are able
to modulate this psychedelic experience
and
you know that the benefit of this is
that oh wow if you're
are doing psychedelics and you're
feeling really overwhelmed move
do a bunch of jumping jacks push-ups
move around space um that's gonna really
help
lower uh the prediction error in your
brain and
uh help you be more uh
back to i guess the reality you're used
to
so yeah this this was a uh
a short uh connection between
psychedelics
and and active in france um
that i just wanted to bring to the space
too
thank you yeah
there was so much um even in that that i
could
uh yeah you can unsure then i will just
um
yep that's perfect i'll just then i'll
just re-crop it
um i guess i'll crop it while i ask this
first question which is
um you pointed out
that a lot of even the clinical work on
psychedelics is done in a way where they
actually prevent movement and also like
education
stay at the desk but don't encourage it

yeah

yeah okay totally yeah like putting

people in not restraint

but it's it's it's only you know

continue with a um authority as a

restraint so just

um yeah what is maybe the role of the

body

in uh remote work or in now that we're

all so

in um again not physically restrained to

our chairs yet but

it feels like that some days it's a

disaster it's

it's um yeah i think we're

uh and i'm a total motor chauvinist

when it comes to our brain i don't know

if you've heard that term before but

no i'm going to explain that yeah so

basically

we have a brain in cognitive functions

because we move

you know when you look at the animal at

the designation between

tree to algae to creatures and the

crazy ocean to animals that move when do

you start getting brains is when they

move

and there's a great example of some

really insane

sea creature that moves around and has a

brain then it finds a place to plop

itself down

and digest its own brain so once

um yeah it doesn't need it our our brain

is needed

to create predictions with the

environment right this is the whole

uh predictive coding and andy clark has
a
beautiful book and papers about about
the feedback groups that we create and
within
with the environment and if our
environment is relatively
highly predictable because we're staying
in the same place
then guess what we don't we
we don't need this uh and we and us
as humans and andy clark ursula goes
into that quite a lot we've sort of
because we want to minimize prediction
error that's that's
part of this whole free energy thing in
our brain
um but we've done it a bit too well
you know we've created these uh large
scale active inferences with
our environment shaping uh our
supermarkets looking the same
as the roads the our house um
and uh i think that's a really a big
cause
of generally the mental health crisis
that we're
seeing and especially when it comes to
then not even moving your body
being in the same state uh physiological
state that's just
true low entropy um
and that's uh really really i think i'm
gonna even
go to the extent of saying like my main
goal and my in my
work um is that one metric
improve the amount of

uh states your body is in for humans
if i can do that then um
for mo you know modern uh citizens that
have a total lack of this
i am absolutely certain that we're gonna
see improvement on
many metrics um that we're lacking
so when it comes uh there and there's
as much as i can to like uh not fall
into
this like i've been working remotely for
uh even before this crazy covert thing
happened
but um you know do calls while you're
walking
outside or stretching outside anything
that doesn't require you to be in front
of a 2d screen
yeah bring toys yeah create
like we do have the ability to uh gamify
and bring toys into our environment
like i have rings outside and punching
bags and
um any anything that will create
constraints around your movements
you'll move your desk change your table
don't work from the
same space if you possibly can just keep
bringing in that prediction
or goes into your body and
um and i think that's a very like
incorrect way
that ergonomics is looked at you know
ergonomics is all about like
let's find the perfect position for you
to be in
for many hours that doesn't exist we're
not supposed to be

in a perfect position just going like
this um
well it's it's really it's like a modern
mental ergonomics and it's the
um ergo the work the dynamics of a
dynamical system
that's us and so the ergonomics is now
the back angle
for the chair but then there's actually
a ratio of the chair and the exercise
ball or whatever that's not my specialty
but it's all about
changing those ratios um because
whatever there is no such thing as
the the perfect sitting state or not
moving state for your body you know
lying down too much
gives us pressure problems
and people in hospital need to be turned
around
to not get these things we are a dynamic
physiological system
and i think there's a very big gap
between people that are studying
these systems a lot of the times and
like math people
and computer science people that are
just really not spending enough time
connecting to their own bodies
um so i think like i highly encourage
everyone to like just find the time to
figure out what you are
as a full human being as like blood
blood pressure bones tendons
um and be in that space and let that
those prediction take precedence at
least
a little bit of your day

yeah one thing you mentioned is this
like top down almost
standardization and and synchronization
and stabilization of the lower level
states
and then the the two affordances we have
are kind of like our physical movement
and our communication
and we've seen how those are
subject to top down as well as bottom up
constraints but that is kind of what's
being
remapped is the constraints on our
movement
variously and our communications
networks variously
and so to have a multi-scale framework
like active inference
where we can talk about the remote team
having input and output and then the
body
we can be like okay here's an algorithm
that's awesome for this
sector for this team but i want to be
not reinforcement learning driven or not
reward driven i want to have a framework
that allows me to express
these kinds of playful states or even
have a rationale
to construct these states and design for
them
yeah so i think there's a you that was a
lot a lot of
good points you brought up so maybe
first the communication thing right
um we're doing this this 2d
sometimes only voice sometimes 2d
screens that none of this is

naturalistic

for our brains right we're 3d creatures

we pick up social cues from being able

to gaze into each other's eyes predict

where you're looking at

um and they and it gets worse the number

of people you

bring in it's just i can't like group

zooms for me are torture there's so much

awkward silence so much inefficiency

who's next who what is the level of even

when we're in a normal social group

you know i usually put little numbers

around people how much i think they

understood me okay this

person understood seventy percent thirty

percent

zero and i do the same for myself and i

try to be very honest with people i like

i understood around thirty percent of

what you just said

um and that's how we are like we have

different uh just data structures

internally

and managing to like minimize the

differences in these data structures

is a very difficult task um

uh even when you're not working remote

and remote work has definitely made

that

a lot more difficult um i've

tried to use things like uh vr

um and found it uh sometimes okay but

also it's

and i'm a superhero you know

enthusiastic person

and still spending more than 15 minutes

in there

at a period of time is just too much
it's heavy it's clunky
it's um a close point of view
um so that's that's a difficult thing
but we do try what we do try to do a lot
of the times is create
um mock-ups directly in vr and
share this the mock-ups in vr so we
understand
what we're trying to build together um
from this 3d perspective
um generally i'm
mew we have like an ar app where i'm
trying to create interactive holograms
so more on the social connection instead
of just
uh getting you on like a 2d
uh screen or the facebook and instagram
but getting something that moves
in my space and actually interacts with
me like you can
blow kisses i don't know if you've ever
seen the roger rabbit movie
but i realized i think i got it like i
started
when i was a kid and then i built this
thing i started the movie again and i
was like oh wow
i just built the roger rabbit thing
kisses through the screen or something
you you blow kisses and it goes into
your 3d space
in ar space so you can actually move
around your space and catch the person's
kisses so
i'm really excited about the possibility
of creating what's called joint
action uh in cognitive neuroscience

remotely and joint action is a
physiological
thing more than anything like if i even
shake your hand or just passing you that
pen is when i start predicting you
and starting to predict you is starting
to build empathy
with you um so these empathy things
really
uh are naturalistically based on very
low level of physiological data
and without this because we're currently
constantly on zoom
and you know i i click here to send you
a button
to send you a chat or i raise my hand by
clicking a thing and you see a thing
over there
it's much much weaker and not
really creating these joint actions so i
think there's a lot to explore
with technology with just joint actions
and um team building activities that are
in vr
even if for a short period of time and
just
simple things like let's build blocks
together
or things that you would do in in
physical space
if you could be together to just
understand
a bit more on this low-level prediction
of
who a person is how they move now vr is
very strange
because you can like mess things up i
don't know if you've ever been in vr

chat

but you can interact with someone as

like oh i'm now

a three meter uh giant with

fire wings and i'm a tiny pixie and

you're just

interacting this way um

so i think that's really interesting and

and

i'm actually it can be a very

interesting way of also de-biasing

things like

oh what happens if we conduct interviews

that are not like we know we know

there's a huge bias

in um in in the whole hiring system

and building teams to begin with that is

not solved there's like

okay you do uh um

you know you don't even write sometimes

hr doesn't look at the names and there's

all these things to try to hide who the

person is but the moment there's

interviews you know who the person is

and

these racial biases come in these gender

biases come in

so there's a lot of really interesting

ways that we could be

de-biasing ourselves by just

again this excessive prediction error

and and the ability to really detach

ourselves from

the normal um perception that we have

um that's really interesting about the

physical components of

remote whether it's a physical object

that's going to be

moving and i don't know like a you know
little jellyfish that does
a movement or just synchronizing and it
made me wonder if there was like
an exercise like could people just say
okay beginning of this video
meeting yeah maybe we all just switched
over a context like
let's do you know three claps and take a
deep breath and just do something
think of a you know purple cow and then
just coordinate that way and then
prepare
or here's our narrative here's what
we're here to do because that kind of
stuff the mission
which comes back to your earlier idea
about like giving
the information in the way that's
naturalistic and what is the natural
unit
it's kind of the the goal the end and so
having the group assemble with the
synchronization and also
a remembrance of the end is like really
an interesting implication i think of
active inference
but not some of these other um sub
component theories
per se yeah and uh so that's also what
i'm
trying to do is this hardware is let
people synchronize their heartbeats
let me you're over there i'm over here
let me send you my heartbeat as
beautiful glowing pulses and you can
send me your heartbeat
and get this uh biophysical data that

honestly if we were even
in the same room together i wouldn't
necessarily get unless i was like
touching you right
so i think we are and this is the
exciting part of like
immersive and embody technology like we
are getting to the stage where we can
start uh moving beyond 2d video
into biophysical data that lets us get a
deeper understanding a deeper connection
and deeper synchronization with each
other
um well
i'm happy to talk for a little bit
longer if you want but i guess my
closing thought or at least direction
because maybe we'll see
or hear from you again would be like
what are the next steps for
applied active inference or for um
an individual who's curious about these
kinds of topics is there
like a website or is there a research
direction where is
active inference coming into play in the
next months to years
yeah so honestly before i heard about
your
podcast i've never heard about any uh
such likes
focus within the predictive coding
framework i would be
very excited about um
connecting and this is something i've
i've been trying to do is bring movers
into the space it seems that there's
these two different worlds

there's movers that hate technology
because they feel it's taking them away
from their bodies
um and there's there's technology and
science and there's a very unfortunately
very small
overlap between them and that's what you
know i was very lucky i found a
co-founder who's
in both these worlds um
but if there was a way to
bridge this gap and like you're saying
these you know
movement exercises it's a thing that
they we do
constantly and improv classes uh
building physiological synchronization
and the things you'll see there
sometimes are just like
this is one organism you know like
people moving together
not pre-decided um and there's
a beautiful research paper i think
no it's
link to it afterwards um but what they
did there
is they tried to figure out what's going
on
uh in the brains of improvisers
um and they have a beautiful like
mathematical
uh paper about the double feedback loop
that's created so it's not and they
tested two conditions leader versus
follower that was more what i was
testing
versus um there is no
leader and follower you're doing things

together

and they found a much higher degree of
synchronization

and a very they built a very cool
paradigm of just like moving your hand
on one up and down that's it very easy
to measure

but it was there the data was there they
managed to find

um a much higher degree of
synchronization when there was no leader
and follower

between um improvisers that were either
musical improvisers or dancers and
trained their brain

to tune in to the other person because
when you're doing this and this is a
very um

you know when you're talking about teams
and synchronizing and
and play i think we really need

a shift in the hierarchical structure of
teams

like i don't see and this is something
that i'm

you know really exploring as a ceo of a
startup

how do i get these ideas that i want out
there

but how do i bring in people's ideas so
they feel that they're part of this

and how do we then get a cohesive
um product at the end of it

not none of these are like easy
questions uh

but having um this ability to

double feedback so it's not just a
top-down manager

right that's saying oh listen this and
this you guys
understand this and this and this this
is my vision
don't expect people to be excited about
your vision
right that's not how these things work
if you want them to be excited
and attuned you need as a you know
manager
in this hierarchical structure be
attuned to their visions what they're
seeing
their perspective why did they join this
project
and try to get that coordinated
um into a global vision that is
um you know doesn't break apart because
there's too many voices
so none of this is easy but i think
that a lot of the times yes going to the
low-level
physiology first and building this
low-level physical
trust with each other and understanding
with each other um
is really unique like me and my
co-founder we dance together we climb on
trees together
before he moved to hawaii now we try to
hang out in vr and do things together
but it's really
not the same um so
um figuring out how to do these joint
physical actions together i think will
really
get us into the future somehow
or a future i want to be part of yeah

nice or
a likely future or one that i expect
myself in um yeah but on that um
tech and movers and sort of the
transdisciplinary
nature and the potentially productive
conversation
uh we we kind of think about these guest
streams like it's too directional
because maybe for those with a physics
background or a philosophy background
you've inspired them to think about how
they're doing their physical
activities or wanting to go deeper into
a knowledge or movement tradition they
were already a part of
and then there's people who that's their
daily life and then maybe this is the
first time they're hearing about active
inference
so we can always do like event series or
discussions or working groups or
events like that or projects like that
to help
make that two-lane freeway
like accessible for everybody and
respect both directions of the
interchange
and it's a really important part like
you said with the moving away from the
hierarchy like of disciplines
you know is physics the top dog
discipline physics is awesome
but it's actually a relationship i
survived
in physics it was dramatic
it's great and then different people
will be drawn to different parts and

that will be sort of like the colony
phenotype
and um when we uh when we
constrain what the lower level looks
like we
increasingly infringe on what the higher
level can be as well
so it's about that balance and so that's
a common theme with these active
inference
discussions well
yeah this is awesome i really appreciate
this conversation and it was uh
i'm sure something that a lot of us will
we listen to and think about so we hope
to maybe
interact with you or other communities
in the future but this was really fun so
thanks a lot sarah
peace bye